

Jinhyung Lee, Ph.D.

Postdoctoral Research Associate, Biomedical Engineering
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RESEARCH AREAS

RNA delivery; DNA-inspired Janus base nanomaterials; nanoparticle assembly and tissue transport; engineered tissue models for cancer and osteoarthritis research.

RESEARCH APPOINTMENTS AND EXPERIENCE

University of Connecticut — Postdoctoral Research Associate | September 2024–present

Department of Biomedical Engineering. Advisor: Yupeng Chen. Research on RNA delivery, Janus base nanomaterials, and engineered tissue models. Co-first author of a 2025 *Journal of Controlled Release* study and co-corresponding author of a 2026 *Materials Today Advances* study.

University of Connecticut — Doctoral research | September 2019–May 2024

Advisor: Yupeng Chen. Developed Janus base nanopieces for intracellular RNA delivery, including a first-author *PNAS* study. Established nucleic-acid delivery protocols and helped build synthesis and characterization infrastructure.

Korea Institute of Science and Technology — Researcher | September 2017–December 2018

Brain Science Institute, Seoul. Developed GPCR-based FRET biosensors for neurotransmitter detection; co-inventor on Korean Patent KR 10-2176538.

University of California, San Diego — Graduate research | January 2016–June 2017

Advisor: Yi Chen. Developed a self-assembling DNA nanogel with embedded functional proteins for intracellular delivery.

EDUCATION

Ph.D., Biomedical Engineering — University of Connecticut | 2024

Advisor: Yupeng Chen. Dissertation: *DNA-Inspired Nanopieces of Drug Delivery in Biomedical Applications*.

M.S., Bioengineering — University of California, San Diego | 2017

Advisor: Yi Chen. Thesis: *Self-Assembly DNA Nanogel with Embedded Functional Proteins for Nanomedicine Platform*.

B.S., Convergence Biotechnology — Konkuk University, Seoul | 2015

College of Bioscience and Biotechnology.

AWARDED RESEARCH FUNDING

ORS/Bethel Fellowship Grant — Principal Investigator | \$20,000

Orthopaedic Research Society / Orland Bethel Family Musculoskeletal Research Center. Awarded; October 2026–September 2027. Co-investigators: Yupeng Chen and Chuan-Ju Liu.

PEER-REVIEWED PUBLICATIONS

Seven publications: six original research articles and one review. Four as first, co-first, or co-corresponding author. † Equal contribution. * Corresponding author.

1. Zhai J†, Surian A†, Cote T, Zhang W, Dang BT, Wang Y, Chen Q, Hoshino K, **Lee J***, Chen Y*. Enhanced endosomal escape for nanoparticle-enabled co-delivery of doxorubicin and siRNA to overcome multidrug resistance. *Materials Today Advances*. 2026;30:100845. doi:10.1016/j.mtadv.2026.100845.
2. **Lee J†**, Zhang W†, Nguyen D, Zhou L, Amengual J, Zhai J, Cote T, Landolina M, Ahmadi E, Sands I, Mishra N, Yu H, Nieh MP, Wang K*, Li Y*, Chen Y*. Computation-aided design of rod-shaped nanoparticles for tumoral targeting. *Journal of Controlled Release*. 2025;387:114169. doi:10.1016/j.jconrel.2025.114169.
3. Zhou L, Zhang W, **Lee J**, Kuhn L, Chen Y*. Controlled self-assembly of DNA-mimicking nanotubes to form a layer-by-layer scaffold for homeostatic tissue constructs. *ACS Applied Materials & Interfaces*. 2021;13(43):51321–51332. doi:10.1021/acsami.1c13345.
4. **Lee J**, Sands I, Zhang W, Zhou L, Chen Y*. DNA-inspired nanomaterials for enhanced endosomal escape. *PNAS*. 2021;118(19):e2104511118. doi:10.1073/pnas.2104511118.
5. Yau A†, **Lee J†**, Chen Y*. Nanomaterials for protein delivery in anticancer applications. *Pharmaceutics*. 2021;13(2):155. doi:10.3390/pharmaceutics13020155. [Review]
6. Sun M, **Lee J**, Chen Y, Hoshino K. Studies of nanoparticle delivery with in vitro bio-engineered microtissues. *Bioactive Materials*. 2020;5(4):924–937. doi:10.1016/j.bioactmat.2020.06.016.
7. Sands I, **Lee J**, Zhang W, Chen Y*. RNA delivery via DNA-inspired Janus base nanotubes for extracellular matrix penetration. *MRS Advances*. 2020;5(16):815–823. doi:10.1557/adv.2020.47.

BOOK CHAPTERS

1. Sands I, **Lee J**, Chen Y. Nanotechnology for DNA and RNA delivery. In: *Nanomedicine: Technologies and Applications*. 2nd ed. Chapter 4. Elsevier; 2023:81–111. doi:10.1016/B978-0-12-818627-5.00008-7.
2. **Lee J**, Chen Y. Nucleic acid delivery systems for gene therapy. In: Zimmer V, ed. *Gene Delivery: Methods and Applications*. Chapter 2. Nova Science Publishers; 2019.

PATENTS

1. Chen Y, **Lee J**, Zhang W. *Therapeutic nanomaterials*. U.S. Patent 12,214,043 B2; issued February 4, 2025. Assignee: University of Connecticut.
2. Chen Y, Zhang W, **Lee J**. *Nanomaterial delivery vehicle and method of use thereof*. U.S. Patent Application Publication 2025/0302762 A1; published October 2, 2025. Application 18/852,925.
3. Seong J, **Lee J**, Kim H, Choo H. *Biosensor based on fluorescence resonance energy transfer for detecting of antipsychotic medicine and detection method for using same*. Korean Patent KR 10-2176538 B1; granted November 9, 2020.

TEACHING AND MENTORING

Teaching Assistant — Foundations of Biomechanics (BENG 226) | April–June 2017

Graduate course, Department of Bioengineering, University of California, San Diego. Led weekly discussion sections, prepared supplementary materials, and graded problem sets and examinations.

Laboratory mentoring — University of Connecticut

Mentored seven graduate and undergraduate researchers in nanoparticle synthesis, characterization, and RNA delivery. Co-authored conference presentations with trainees at BMES and SCCUR. Helped establish laboratory workflows as the founding graduate student and trained researchers who joined later.

SELECTED PRESENTATIONS

Invited talks

1. DNA-inspired Janus base nanoparticles for drug delivery in biomedical applications. Chonnam National University Medical School Seminar, Republic of Korea, 2026.
2. Computation-aided design of rod-shaped Janus base nanotubes for enhanced tumoral targeting. Biomaterials and Biofabrication Conference, USA, 2025.
3. DNA-inspired nanopieces of drug delivery in biomedical applications. College of Medicine, Kyung Hee University Seminar, Republic of Korea, 2024.
4. DNA-inspired nanomaterials for small molecule delivery to hard-to-penetrate solid tumors. Next-Generation Nanomedicine, USA, 2023. Short talk.

Podium presentation

Janus base nanopieces delivered mRNA therapeutics against cartilage degradation and joint pain for osteoarthritis treatment. Orthopaedic Research Society Annual Meeting, 2024.

AWARDS AND RECOGNITION

ORS/Bethel Fellowship Grant, Orthopaedic Research Society, 2026; \$20,000.
Predoctoral Fellowship, University of Connecticut, 2023.
RNATx Young Investigator Travel Award, UMass Chan Medical School, 2023.
Graduate Travel Awards, University of Connecticut; four competitive awards, 2020–2022.
Sung-jo Merit Scholarship and Shin-jo Merit Scholarship, Konkuk University, 2014–2015.

PROFESSIONAL SERVICE AND MEMBERSHIPS

Reviewer, *ASSAY and Drug Development Technologies*, 2026.
Member, Orthopaedic Research Society.
Member, American College of Rheumatology.
Member, Biomedical Engineering Society.